The generative model of sec. 5.10, the learning operators of sec. 5.15, and the corruption process of sec. 5.19 are exercised by 9 studies, including the contaminated-sentinel robustness sweep (Study 4). The shared configuration (seed budget, colony size, contamination rates, divergences, trial counts, and the statistics settings) is read from `experiment:` in `manuscript/config.yaml`; the remaining per-study parameters are tested code defaults in `src/fedference/experiments/`. No value is hard-coded in the manuscript, and each token below resolves to the same configuration the code executed.

Determinism through fixed seeds and generated variables I

All stochastic steps draw from explicitly seeded generators (`np.random.default_rng`); the global `np.random` state is never touched. The single-run studies use the first configured seed (0), and the across-seed studies enumerate the deterministic seed list $0, \dots, n_{\mathsf{seeds}} - 1$. Re-running with the same seed reproduces every number in the results bit-for-bit, so the bootstrap confidence intervals and paired-test p-values of sec. 5.27 are themselves deterministic functions of the seed.

Determinism through fixed seeds and generated variables I

The figure layer follows the same provenance rule. Captions are written to be self-contained, with axes, resampling units, deterministic runs, truncated axes, and error-band status disclosed in the caption rather than left to inference (Rougier et al. 2014; Midway 2020). This is why several results figures state "single deterministic run" or "no error band" even when the inferential evidence appears in an adjacent table.

Studies 1–3 implement reduced categorical protocols that are source-mechanism analogues of the belief-sharing, language-acquisition, and model-reduction mechanisms discussed by Friston et al. (2024) on the sentinel POMDP; they are not exact source-protocol figure replications. The sweep adds the FedGVI robustness contribution (Mildner et al. 2025). Unless a study specifies otherwise, the global defaults are $n_{\text{seeds}} = 480$ independent seeds and $n_{\text{trials}} = 960$ matched trials per condition.

Study suite and contamination sweep I

Table 1: Per-study configuration, read from `experiment:` in `config.yaml` and surfaced as manuscript tokens by `src/manuscript_variables.generate_variables`. The sample sizes carried into the statistics — the across-seed belief-sharing sample ($n = 480$ seeds), the language trajectory (25 ordered points summarized over $n = 480$ independent seeds), and the paired robustness trials ($n = 960$ per condition) — are reported with their respective results.

Study	What it measures	Key parameters
1 — Belief sharing	Communicating vs. incommunicado colony free energy	`n_agents = 7`, `acuity = 0.55`
2 — Language acquisition	Dirichlet-learning KL descent (eq. 15)	`num_steps = 24`
3 — Emergence / BMR	Reduced-vs-full model evidence (eq. 16)	candidate states `n = 4`
4 — Robustness sweep	Robust vs. naive consensus under contamination	`n_agents = 7`, `n_contaminated = 2`
5 — Disjoint-FOV moving world	Communication necessity with non-overlapping fields of view	`n_agents = 3`, `fov_width = 2` (Supplement, sec. 13.7)

Study 1 — belief sharing. A colony of 7 sentinels at the deliberately low acuity 0.55 each infer the creature location (eq. 12) and share beliefs through the log-linear pool (eq. 6). We compare a *communicating* colony against an *incommunicado* one of the same size and seed, scoring each by the mean variational free energy of eq. 13. Two protocol details: state inference in this study substitutes a flat (uniform) prior for D, and each belief's free energy is scored against the pooled evidence of all agents' observations (the disclosure carried in sec. 7.2). The across-seed sample is $n = 480$ seeds, one colony-mean free energy per seed (fig. 8, fig. 9).

Study 2 — language acquisition. Each configured seed runs one sentinel trajectory using the conjugate Dirichlet update (eq. 14) over 24 steps. We record the KL descent of eq. 15 at each ordered step, giving 25 points per trajectory and $n = 480$ independent seed trajectories for the pointwise interval in fig. 10.

Study 3 — emergence. A redundant model reduction and a supported one are scored by the BMR free energy of eq. 16 over $n = 4$ candidate states (fig. 11).

Study 4 — robustness sweep. The sweep varies two factors on a colony of 7 sentinels of which 2 are saboteurs (`confident_wrong`, sec. 5.20):

Study suite and contamination sweep I

- **Contamination rate** over $\{0, 0.225, 0.45, 0.675, 0.9\}$ — the convex-mix weight of eq. 17 toward the confident-wrong delta.
- **Server robustness setting**, named with FedGVI client-loss/ divergence vocabulary for cross-reference only $\{KLD, RKL, AR, beta, rcce\}$. `KLD` is the non-robust Friston / standard-Bayes baseline (server robustness 0, eq. 7) and serves as the design's negative control: the recovery identity guarantees it reproduces the naive log-linear pool exactly, so every robust-versus-naive contrast is scored against a comparator that is provably the un-robustified server rather than a separately tuned competitor. The remaining labels $\{RKL, AR, beta, rcce\}$ each select a fixed `robust_aggregate` down-weighting constant (the executed mapping is KLD (c=0.00), RKL (c=1.50), AR (c=1.30), beta (c=1.70), rcce (c=1.60); defined in `fedference.experiments._common`). None of these labels invoke the client-side `generalized_posterior` update of sec. 5.3 or the divergence family of sec. 5.6; this sweep therefore exercises only the server-side heuristic axis of sec. 3.3 (`robust_aggregate`), never the per-agent rigorous axis. The executed per-divergence down-weighting strengths are fixed constants defined in `fedference.experiments._common` and recorded in the run reports.

The headline verdict pairs 960 independent trials at the fixed contamination rate 0.800 — heavy contamination that degrades the naive pool while staying below the pure-veto cliff (fig. 12, fig. 13). Each trial contributes one matched (naive, robust) accuracy pair, so the replication unit is the trial and the estimand is the within-trial accuracy difference at that single rate. A complementary federated logistic-regression baseline applies the same client-side robust loss to flipped-label contamination, isolating the rigorous axis (fig. 16).

Study 5 — disjoint-FOV moving world. This extension (sec. 13.7) places 3 agents on a 2-slot disjoint-FOV track to test the necessity of communication when agents cannot observe the same positions. Isolated agents are compared against EFE-guided communicating agents on accuracy across the moving sentinel's trajectory. Five structural extension studies (Studies 5–9, Supplementary sections) build on the same POMDP substrate and are described there: the moving disjoint-FOV sentinel (sec. 13.7), the 2-level hierarchical POMDP (sec. 13.9), the N-level extension (sec. 13.11), the 2-D sensitivity sweep (sec. 13.13), and parameter recovery (sec. 14).

The verdict design answers a deliberate question: pair *many* trials at *one* high contamination rate rather than spread few trials across the rate curve. A matched-pairs Wilcoxon test gains power from the number of matched pairs, so concentrating $n = 960$ paired trials at the single rate 0.800 gives the test the resolution to detect the robustness effect; scattering the same budget across 5 rates would dilute every contrast. The across-seed studies are powered separately, with $n = 480$ seeds for belief sharing and $n = 480$ independent seed trajectories for language acquisition. The 25 ordered learning points are repeated measures within each seed, not additional independent replicates, so the language interval does not count time points as samples. The structural-extension and cross-study summary tier uses $n = 128$ independent seeds and $n = 40$ matched trials per contamination rate for its robustness row. Trials and clients are nested within a seed and are reduced before across-seed inference; they are not additional independent replicates.

The bounded red-team review grid is a separate source-bound analysis profile. It uses 160 deterministic seed replicates, with 24 trials nested within each seed and scenario/rate cell, and retains the registered rates 0, 0.2, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9 across the finite attack union clean, confident wrong, permutation, byzantine, drift, label noise, uniform. Its independent unit is the configured seed within a declared cell; cells that share design structure are not treated as independent worlds. Robust operating points, method settings, and rate profiles are fixed before the review run. The grid reports selection-free contrasts and keeps clean, uniform, label-noise, permutation, confident-wrong, Byzantine, and drift controls visible.

Completion of this finite review does not close server-heuristic characterization, leakage-free calibration, external-data, or protocol-reconstruction phases.

We do not merely assert adequacy — we compute the design power implied by the observed effect. For the headline robust method (RKL) the observed-effect design power of the paired Wilcoxon at the run's $n = 960$, computed at $\alpha = 0.05$ against the directional alternative `greater` (robust accuracy exceeds naive), is 1.0000. The power approximation uses the deterministic noncentral-normal approximation of sec. 5.27, deflated by the Wilcoxon's Pitman asymptotic relative efficiency, so it is approximately calibrated for the signed-rank test actually run (exact under a normal-shift alternative; the power computation is one-sided while the reported p-values are two-sided). To bound a confirmatory replication, the prospective sample size needed to reach the target power 0.80 at the observed effect is $n = 5$ matched trials — the explicit sample-size budget a follow-up study should adopt. These power quantities characterize the *server-side* aggregation contrast; per the honesty contract of sec. 3.3 they do not certify the per-client β/rcce guarantees, which are pinned by the locked core (sec. 6) rather than by these aggregation-level statistics.

The FedGVI core, the POMDP studies, and the logistic-regression baseline are pure NumPy / SciPy — no GPU and no network. The deterministic-MLP neural complement (sec. 7.6) additionally uses PyTorch (CPU); the analysis pipeline executes it and emits its numbers as tokens exactly like every other result when the `torch` optional extra is installed, and otherwise records a skipped status with unavailable-value sentinels instead of silently fabricating neural results. Versions and platform — including the PyTorch version used for the MLP complement — are recorded automatically in sec. 10.